1	$x^n \to x^{n+1}$ $\int\left(8x^3 - 6\sqrt{x} - \frac{2}{5x^3}\right)\mathrm{d}x = 2x^4 - 4x^{\frac{3}{2}} + \frac{1}{5x^2} + c$	M1 A1A1A1
		(4 marks)

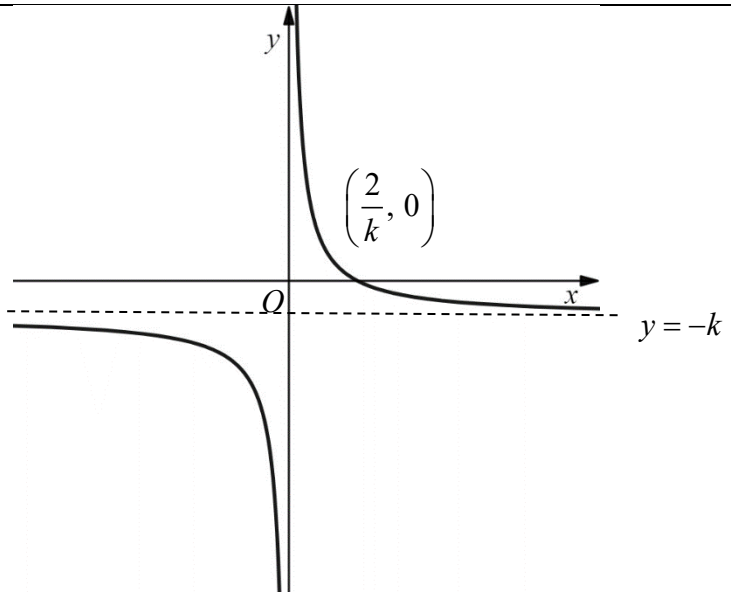
2(a)	Gradient $= \dfrac{8-5}{7\sqrt{3}-\left(-2\sqrt{3}\right)}$ o.e.	M1
	e.g. $\Rightarrow \dfrac{3}{9\sqrt{3}} \times \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{\sqrt{3}}{9}$	A1
		(2)
(b)	$\dfrac{1}{9}\sqrt{3} \rightarrow -\dfrac{9}{\sqrt{3}}$	B1ft
	$y-5 = "-\dfrac{9}{\sqrt{3}}"\left(x+2\sqrt{3}\right)$	M1
	$y = -3\sqrt{3}x - 13$	A1
		(3)
		(5 marks)

3(a)	$58^2 = a + 5^3 \times b$, $65^2 = a + 10^3 \times b \Rightarrow a = ...$, or $b = ...$	M1
	$a = 3241,\ b = 0.984$	A1A1
		(3)
(b)	e.g. $85^2 = "3241" + "0.984" \times T^3 \Rightarrow T = ...$	M1
	$(T =)$ awrt 15.9	A1
		(2)
		(5 marks)

4(i)(a)	ay^3	B1
		(1)
(i)(b)	$\dfrac{5}{\left(3a^{1-x}\right)^{-2}} = \dfrac{45}{\ldots}$	B1
	$\left(\ldots a^{1-x}\right)^{-2} = a^{-2} \times a^{2x}$	M1
	$= \dfrac{45a^2}{y^2}$	A1
		(3)
(ii)(a)	e.g. $3^{4t+2} = \left(3^{2t}\right)^2 \times 9 = 9p^2$	M1
	$\Rightarrow 27 \times p^2 + 3 = 82 \times p \Rightarrow 27p^2 - 82p + 3 = 0$ *	A1*
		(2)
(ii)(b)	Solves the quadratic $\Rightarrow$ 3, $\dfrac{1}{27}$	B1
	$\left(3^t\right)^2 = 3$ or $9^t = 3 \Rightarrow t = \dfrac{1}{2}$	M1
	or	
	$\left(3^t\right)^2 = \dfrac{1}{27} \Rightarrow 3^t = \dfrac{1}{3\sqrt{3}} \Rightarrow t = -\dfrac{3}{2}$ or $9^t = \dfrac{1}{27} \Rightarrow t = -\dfrac{3}{2}$	
	$(t =) \dfrac{1}{2}, -\dfrac{3}{2}$	A1
		(3)
		(9 marks)

5(a)	$x^n \to x^{n-1}$	M1
	$\left(\frac{dy}{dx} =\right) 12x^2 - \frac{2}{x^2}$	A1
		(2)
(b)	$12x^2 - \frac{2}{x^2} = -5 \Rightarrow ... \Rightarrow 12x^4 + 5x^2 - 2 = 0$ *	M1A1*
		(2)
(c)	$12x^4 + 5x^2 - 2 = 0 \Rightarrow (4x^2 - 1)(3x^2 + 2) = 0 \Rightarrow x^2 = \frac{1}{4} \to x = \frac{1}{2} \quad (x > 0)$	M1
	$y = 4\left(\frac{1}{2}\right)^3 + \frac{2}{\left(\frac{1}{2}\right)} + 9 = ... \left(= \frac{27}{2} \right)$	M1
	$k = "\frac{27}{2}" + 5 \times "\frac{1}{2}" = ...$	dM1
	$k = 16$	A1
		(4)
		(8 marks)

6(a)	$\dfrac{2(4)^2+a\times4+b}{4\sqrt{4}}=7\Rightarrow32+4a+b=56\Rightarrow4a+b=24$ *	M1A1*
		(2)
(b)	$4a+b=24,\ a+b=-9\Rightarrow a=11,\ b=-20$	M1A1
	$\dfrac{x^{\frac{3}{2}}}{2}+\dfrac{11x^{\frac{1}{2}}}{4}-5x^{-\frac{1}{2}}$	M1
	$\int\dfrac{x^{\frac{3}{2}}}{2}+\dfrac{11x^{\frac{1}{2}}}{4}-5x^{-\frac{1}{2}}\,\mathrm{d}x\Rightarrow$ Two of $\dfrac{x^{\frac{5}{2}}}{5},\ \dfrac{"11"x^{\frac{3}{2}}}{6},\ \dfrac{"-20"}{2}x^{\frac{1}{2}}$	dM1A1ft
	$\dfrac{(4)^{\frac{5}{2}}}{5}+\dfrac{11(4)^{\frac{3}{2}}}{6}-10(4)^{\frac{1}{2}}+c=-5\Rightarrow c=...$	M1
	$\left(\mathrm{f}(x)=\right)\ \dfrac{1}{5}x^{\frac{5}{2}}+\dfrac{11}{6}x^{\frac{3}{2}}-10x^{\frac{1}{2}}-\dfrac{91}{15}$	A1
		(7)
(c)	$(7,\ -5)$	B1
		(1)
		(10 marks)

7(a)		B1B1B1
		(3)
(b)	$-kx-6=\frac{2}{x}-k \Rightarrow -kx^2-6x=2-kx \Rightarrow kx^2+(6-k)x+2=0$ $(6-k)^2-4\times k\times 2 \Rightarrow k^2-20k+36$ $\Rightarrow \text{CVs}=2,18 \Rightarrow k<"2" \text{ or } k>"18"$ $(0<)k<2 \text{ or } k>18$	M1 dM1A1 M1 A1
		(5)
		(8 marks)

8(a)	$\frac{\sin OAB}{3.4} = \frac{\sin\left(\frac{\pi}{6}\right)}{1.9} \Rightarrow OAB = 2.034$	M1A1A1
		(3)
(b)	$\text{Area} = \frac{1}{2} \times 3.4 \times 1.9 \times \sin\left(\pi - \frac{\pi}{6} - "2.034"\right) = \text{awrt}\, 1.78 \quad (\text{cm}^2)$	M1A1
		(2)
(c)	Area of sector $OBC = "1.78" \times \frac{3}{2}$ $(= "2.67")$	M1
	$\frac{1}{2} \times 3.4^2 \times \theta = "\frac{1.78 \times 3}{2}"$	M1
	$\theta = \text{awrt}\, 0.462$ *	A1*
		(3)
(d)	$OA = \sqrt{3.4^2 + 1.9^2 - 2 \times 3.4 \times 1.9 \times \cos\left(\pi - \frac{\pi}{6} - "2.034"\right)}$	M1
	$OA = \text{awrt}\, 2.1 \ (\text{cm})$	A1
	Length of arc $BC = 0.462 \times 3.4$ $(= 1.57...)$	B1
	Perimeter $= 3.4 + 1.9 + "2.1" + 0.462 \times 3.4$	dM1
	$= \text{awrt}\ 9\,(\text{cm})$	A1
		(5)
		(13 marks)

9(a)	$b=-1$	B1
	$\frac{9}{4}-(x-3)^2$	M1A1
		(3)
(b)	$\left(3, \frac{9}{4}\right)$	B1ftB1ft
		(2)
(c)(i)	$x=\frac{9}{2}$	B1
(ii)	$k=\frac{\pi}{3}$	B1
(iii)	6	B1
		(3)
(d)	$m=\frac{0-"\frac{9}{4}"}{"\frac{9}{2}"-"3"}=-\frac{3}{2}$	M1
	$y-0="-\frac{3}{2}"\left(x-"\frac{9}{2}"\right)$ or $y-"\frac{9}{4}"="-\frac{3}{2}"(x-"3")$	dM1
	$y=-\frac{3}{2}\left(x-\frac{9}{2}\right)$ or $y-\frac{9}{4}=-\frac{3}{2}(x-3)$ o.e.	A1
	$y>\cos\left(\frac{1}{3}\pi x\right),\ y<6x-\frac{27}{4}-x^2,\ y<-\frac{3}{2}x+\frac{27}{4}$	B1ftB1
		(5)
		(13 marks)